🧪 Lab: Python + MySQL Interaction (Invoice System)

🎯 Objective

In this lab, you will learn how to:

- Create a MySQL database
- Insert sample data
- Connect Python to MySQL
- Use dynamic SQL queries
- Generate a simple invoice from database data

🧠 Concept (Simple Understanding)

Think of this system like a store 🏪:

- MySQL = storage (stores orders, customers, products)
- Python = cashier (asks for order and prints receipt)
- User = customer entering order number

🧰 Step 1: Open Terminal

Open your terminal and log into MySQL:

```
mysql -u root -p
```

⚠️ IMPORTANT NOTE (READ FIRST)

👉 You must use a **valid MySQL username and password** in your Python code.

- If your MySQL requires a password → include it in Python
- If you created a new user (recommended), use that account

Example:

```
user="xian"
password="1234"
```

❗ If you forget to include your correct password, you will get:

```
Access denied for user 'root'@'localhost'
```

🧰 Step 2: Create Database

```sql
CREATE DATABASE demo_db;
USE demo_db;
```

🧰 Step 3: Create Tables

```sql
CREATE TABLE customer (
    id INT PRIMARY KEY,
    name VARCHAR(50),
    zip VARCHAR(10)
);

CREATE TABLE orders (
    order_id INT PRIMARY KEY,
    customer_id INT,
    order_date DATE
);

CREATE TABLE product (
    product_id INT PRIMARY KEY,
    name VARCHAR(50),
    price FLOAT
);

CREATE TABLE order_item (
    order_id INT,
    product_id INT,
    quantity INT
);
```

🧰 Step 4: Insert Sample Data

```sql
INSERT INTO customer VALUES (1, 'Alice', '02115');
INSERT INTO customer VALUES (2, 'Bob', '10001');

INSERT INTO orders VALUES (101, 1, '2024-01-01');
INSERT INTO orders VALUES (102, 2, '2024-01-02');

INSERT INTO product VALUES (1, 'Pen', 2.5);
INSERT INTO product VALUES (2, 'Book', 10.0);
```

```
INSERT INTO order_item VALUES (101, 1, 2);
INSERT INTO order_item VALUES (101, 2, 1);
INSERT INTO order_item VALUES (102, 2, 3);
```

🧰 Step 5: Verify Data

```
SELECT * FROM customer;
SELECT * FROM orders;
SELECT * FROM product;
SELECT * FROM order_item;
```

🧰 Step 6: Test SQL Query

```
SELECT c.name, c.zip,
       o.order_date,
       p.name, oi.quantity, p.price
FROM orders o
JOIN customer c ON o.customer_id = c.id
JOIN order_item oi ON o.order_id = oi.order_id
JOIN product p ON oi.product_id = p.product_id;
```

🧰 Step 7: Exit MySQL

```
exit;
```

🧰 Step 8: Install Python Connector

```
pip install mysql-connector-python
```

🧰 Step 9: Create Python File

```
nano demo_invoice.py
```

🧰 Step 10: Write Python Code

```
import mysql.connector

conn = mysql.connector.connect(
    host="localhost",
    user="root",      # use your own MySQL username
```

```
    password="11111",  # use your own password
    database="demo_db" #use your created database name
)

cursor = conn.cursor()

order_id = input("Enter Order ID: ")

query = """
SELECT c.name, c.zip,
       o.order_date,
       p.name, oi.quantity, p.price
FROM orders o
JOIN customer c ON o.customer_id = c.id
JOIN order_item oi ON o.order_id = oi.order_id
JOIN product p ON oi.product_id = p.product_id
WHERE o.order_id = %s
"""

cursor.execute(query, (order_id,))
rows = cursor.fetchall()

total = 0

print("\n===== SIMPLE INVOICE =====")

for row in rows:
    name, zip_code, date, product, qty, price = row
    amount = qty * price
    total += amount

    print(f"{name} | {product} | {qty} x {price} = {amount}")

print(f"TOTAL: {total}")

cursor.close()
conn.close()
```

🧰 Step 11: Save File

- Press Ctrl + O
- Press Enter
- Press Ctrl + X

🧰 Step 12: Run Program

python demo_invoice.py

🧰 Step 13: Enter Order ID

Example:

101

✅ Expected Output

===== SIMPLE INVOICE =====
Alice | Pen | 2 x 2.5 = 5.0
Alice | Book | 1 x 10.0 = 10.0
TOTAL: 15.0

🧠 Key Learning Points

- Python connects to MySQL using credentials
- SQL query retrieves data
- JOIN combines multiple tables
- Dynamic SQL allows user input
- Python formats results into readable output

⚠️ Common Errors

Error	Cause	Fix
Access denied	Missing/wrong password	Add correct credentials
Unknown column	Wrong column name	Check with DESCRIBE
No results	Order ID not exist	Check data

🎯 Summary

User input → Python → SQL → Database → Output

Submit:

demo_db.sql
demo_invoice.py
screenshot.png